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Principio de Inclusión-Exclusión

Proposición 1 (Principio de Inclusión-Exclusión). *Sea* (X, Σ, μ) *un espacio de medida y sean* $A_1, \ldots, A_n \in \Sigma$

$$\implies \mu\left(\bigcup_{j\in\mathbb{N}_n} A_j\right) = \sum_{I\subset\mathbb{N}_n} (-1)^{|I|+1} \mu\left(\bigcap_{j\in I} A_j\right)$$

***Demostración*:** Observamos que para $n = 1$ se cumple trivialmente. Para $n = 2$ podemos expresar $A_1 \cup A_2$ como la unión de sus partes disjuntas: $A_1 \setminus A_2$, $A_1 \cap A_2$ y $A_2 \setminus A_1$

$$\begin{aligned}\implies \mu\left(A_1 \cup A_2\right) + \mu\left(A_1 \cap A_2\right) &= \mu(A_1 \setminus A_2) + \mu(A_2 \setminus A_1) + 2\mu(A_1 \cap A_2)\\ &= \mu\left(A_1 \setminus A_2 \sqcup A_1 \cap A_2\right) + \mu\left(A_2 \setminus A_1 \sqcup A_1 \cap A_2\right)\\ &= \mu(A_1) + \mu(A_2).\end{aligned}$$

Luego despejando obtenemos $\mu\left(A_1 \cup A_2\right) = \mu(A_1) + \mu(A_2) - \mu(A_1 \cap A_2)$.

Supongamos por inducción que se cumple para $n \in \mathbb{N}$ arbitrario, entonces para $n + 1$:

$$\begin{aligned}\mu\left(\bigcup_{j=1}^{n+1} A_j\right) = \mu\left(A_{n+1} \cup \bigcup_{j=1}^{n} A_j\right) &\overset{\substack{\text{caso } n=2\\ \downarrow}}{=} \mu(A_{n+1}) + \mu\left(\bigcup_{j=1}^{n} A_j\right) - \mu\left(A_{n+1} \cap \bigcup_{j=1}^{n} A_j\right)\\ &\overset{\substack{\text{caso inductivo}\\ \downarrow}}{=} \mu\left(A_{n+1}\right) + \sum_{I\subset\mathbb{N}_n} (-1)^{|I|+1} \mu\left(\bigcap_{j\in I} A_j\right) - \mu\left(A_{n+1} \cap \bigcup_{j=1}^{n} A_j\right)\\ &= \sum_{I\subset\mathbb{N}_{n+1}} (-1)^{|I|+1} \mu\left(\bigcap_{j\in I} A_j\right).\end{aligned}$$

$\blacksquare$

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